

The Hubble Constant in the T0 Model: Energy Loss Through the Universal ξ -Field

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Abstract

The T0 model interprets cosmological redshift not as expansion but as energy loss of photons in the universal ξ -field. This document treats the field-theoretic energy loss mechanism and its connection to the Hubble constant. The scale of the redshift is calibrated via the measured H_0 (P39); as structural form the corpus carries $\hbar H_0/(m_e c^2) = \frac{\pi}{2}\xi^{10}$ with $H_0 \approx 66.82$ km/s/Mpc (Doc. 165, 388). Unit conversion, frequency independence, and the Hubble tension are treated in Document 026 (Geometric Cosmology).

.1 Introduction: Rethinking the Hubble Constant

The conventional interpretation of Hubble's law assumes galaxies recede due to expanding space ($v = H_0 d$). This expansion paradigm requires 69% dark energy, suffers from persistent measurement tensions, and fine-tuning problems.

The T0 model offers a fundamentally different perspective: the universe is static, and the observed redshift arises from energy loss of photons as they propagate through the universal ξ -field. The Hubble constant thus transforms from a measure of space expansion into a characteristic energy loss rate.

The time-energy duality ($\Delta E \cdot \Delta t \geq \hbar/2$) forbids a temporal beginning of the universe: a finite time interval from a singularity would require infinite energy uncertainty. The universe must have existed eternally, making space expansion unnecessary for explaining cosmic observations.

.2 The Universal ξ -Field

The cornerstone of the T0 model is the universal geometric constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333... \times 10^{-4} \quad (1)$$

This dimensionless constant determines a characteristic energy scale:

$$E_\xi = \frac{1}{\xi} = 7500 \quad (\text{natural units}) \quad (2)$$

The ξ -field permeates all of space and mediates interactions between photons and the vacuum. Its dynamics are described by the wave equation:

$$\square E_{\text{field}} = \left(\nabla^2 - \frac{\partial^2}{\partial t^2} \right) E_{\text{field}} = 0 \quad (3)$$

.3 Energy Loss Mechanism

Fundamental Energy Loss Equation

Photons lose energy through interaction with the ξ -field. The rate of energy loss follows the differential equation:

$$\frac{dE}{dx} = -\frac{H_0}{c} \cdot E \quad (4)$$

The negative sign indicates energy loss; the linear dependence on E ensures that the resulting redshift is frequency-independent (see Document 026 for the geometric justification). The prefactor must be an inverse length: with the dimensionless ξ alone, $-\xi E$ would have dimension $[E]$, whereas dE/dx in natural units has $[E^2]$; likewise $\xi \cdot x$ would have the dimension of a length $[E^{-1}]$. With x in metres, $z = \xi x$ would already reach $z = 1$ at $x = 1/\xi = 7500$ m. The equations are therefore written with the inverse length H_0/c .

Solution for Cosmological Distances

For cosmological observations ($H_0 x/c \ll 1$), the perturbative solution yields:

$$E(x) = E_0 \left(1 - \frac{H_0}{c} \cdot x \right) \quad (5)$$

The redshift is therefore:

$$z = \frac{E_0 - E(x)}{E(x)} \approx \frac{H_0 x}{c} \quad (6)$$

This relationship reproduces the observed linear Hubble law without space expansion.

.4 Connection to the Hubble Constant

Comparison with Hubble's law $z = H_0 d/c$ fixes the scale of the energy loss only via the measured H_0 ; in this sense H_0 is calibrated, not derived from ξ alone (P39). As a structural relation the corpus carries (Doc. 165):

$$H_0^{\text{T0}} = \frac{\pi}{2} \xi^{10} \frac{m_e c^2}{\hbar} \quad (7)$$

Result: $H_0^{\text{T0}} \approx 66.82$ km/s/Mpc (-0.86% relative to Planck). The exponent 10 and the prefactor $\pi/2$ are not derived.

The older form $H_0 = E_H/\hbar$ with $E_H = E_0 \xi^{41/4}$ (≈ 66.2 km/s/Mpc) does not hold: via the relation $T_{\text{CMB}}^4 = 16 H_0 m_e^3$, carried as a candidate in Doc. 388, it gives $T_{\text{CMB}} = 2.71898$ K, 11σ off the FIRAS value, while the form with exponent 10 agrees to 0.1σ .

For the step-by-step unit conversion (natural units $\rightarrow$ SI $\rightarrow$ km/s/Mpc), comparison with experimental measurements, and the Hubble tension, see **Document 026** (Geometric Cosmology, Sections 4–6).

.5 Dimensional Analysis

Verification of dimensional consistency in natural units:

Quantity	T0 Expression	Dimension	Status
Geometric constant	$\xi = 4/3 \times 10^{-4}$	[1]	✓
Energy scale	$E_\xi = 1/\xi$	[E]	✓
Energy loss rate	$dE/dx = -(H_0/c) \cdot E$	[E ²]	✓
Redshift	$z = H_0 x/c$	[1]	✓
Hubble parameter	$H_0 = \frac{\pi}{2} \xi^{10} m_e c^2 / \hbar$	[T ⁻¹]	✓

The rows energy loss rate and redshift are consistent only with the inverse length H_0/c ; the form with ξ alone ($-\xi \cdot E$ with dimension [E], $\xi \cdot x$ with dimension [E⁻¹]) does not hold (see the section on the energy loss mechanism).

.6 Theoretical Advantages

The reinterpretation of the Hubble constant as an energy loss rate solves several problems:

1. **Elimination of dark energy:** The apparent cosmic acceleration arises naturally from distance-dependent energy loss. No exotic energy form with negative pressure is required.
2. **Fine-tuning problems solved:** The horizon problem vanishes since all regions have been in causal contact over infinite time. The flatness problem disappears for lack of initial conditions.
3. **Parameter reduction:** Where Λ CDM requires six independent parameters, T0 traces the cosmological observables back to the geometric constant ξ ; the scale of the redshift is fixed by the measured H_0 and does not follow from ξ alone.

.7 Conclusion

$$H_0^{T0} = \frac{\pi}{2} \xi^{10} \frac{m_e c^2}{\hbar} \approx 66.82 \text{ km/s/Mpc} \quad (8)$$

The universe is not expanding. The Hubble constant measures energy loss in the ξ -field, not recession; its scale is calibrated via the measured value (P39), and the structural form with exponent 10 is supported by the CMB temperature (Doc. 388). The finite element simulation of the T0 vacuum, frequency independence as geometric proof, and the Hubble tension are found in Document 026.

Bibliography

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